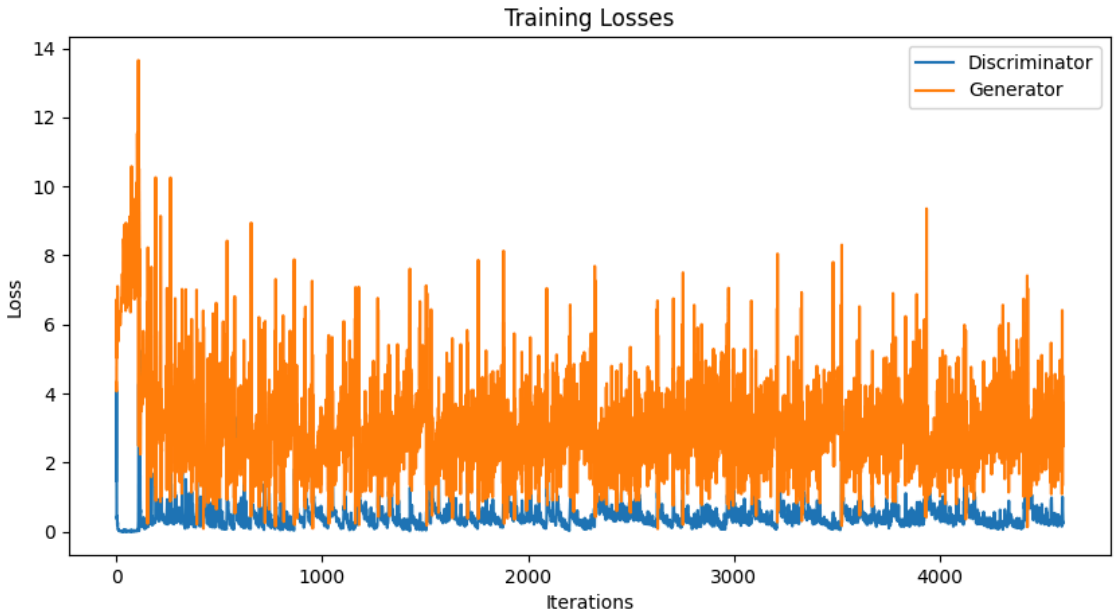
实验报告——用pytorch实现MNIST上DCGAN的训练

训练过程观察到的现象

总结来说，`d_loss` `g_loss`呈现交替震荡但振幅逐渐缩小的趋势。

- `d_loss` `g_loss`都不是单调下降的，往往`d_loss`变小了`g_loss`就会变大
- `d_loss` `g_loss`早期迅速下降，之后趋于平稳



```
Using device:  cuda:0
[Epoch 0/10] [Batch 0/469] D_loss: 1.4825 G_loss: 4.3198
[Epoch 0/10] [Batch 100/469] D_loss: 0.0420 G_loss: 10.1217
[Epoch 0/10] [Batch 200/469] D_loss: 0.3653 G_loss: 2.9907
[Epoch 0/10] [Batch 300/469] D_loss: 0.3352 G_loss: 3.0448
[Epoch 0/10] [Batch 400/469] D_loss: 0.3442 G_loss: 2.5166
Epoch 0 took 52.49 seconds
[Epoch 1/10] [Batch 31/469] D_loss: 0.1226 G_loss: 3.6722
[Epoch 1/10] [Batch 131/469] D_loss: 0.3155 G_loss: 2.1993
[Epoch 1/10] [Batch 231/469] D_loss: 0.5076 G_loss: 2.7789
[Epoch 1/10] [Batch 331/469] D_loss: 0.6595 G_loss: 3.5258
[Epoch 1/10] [Batch 431/469] D_loss: 0.4075 G_loss: 1.3189
Epoch 1 took 52.05 seconds
[Epoch 2/10] [Batch 62/469] D_loss: 0.3006 G_loss: 2.4973
[Epoch 2/10] [Batch 162/469] D_loss: 0.1181 G_loss: 3.9229
[Epoch 2/10] [Batch 262/469] D_loss: 0.5909 G_loss: 2.5563
[Epoch 2/10] [Batch 362/469] D_loss: 0.2941 G_loss: 2.1421
[Epoch 2/10] [Batch 462/469] D_loss: 0.0899 G_loss: 3.1571
Epoch 2 took 52.35 seconds
[Epoch 3/10] [Batch 93/469] D_loss: 0.0998 G_loss: 4.2411
[Epoch 3/10] [Batch 193/469] D_loss: 0.4567 G_loss: 3.3998
[Epoch 3/10] [Batch 293/469] D_loss: 0.1894 G_loss: 3.9774
[Epoch 3/10] [Batch 393/469] D_loss: 0.4476 G_loss: 2.8839
Epoch 3 took 52.24 seconds
[Epoch 4/10] [Batch 24/469] D_loss: 1.2067 G_loss: 0.3835
[Epoch 4/10] [Batch 124/469] D_loss: 0.5923 G_loss: 1.6440
[Epoch 4/10] [Batch 224/469] D_loss: 0.7400 G_loss: 2.2002
[Epoch 4/10] [Batch 324/469] D_loss: 0.0248 G_loss: 5.3878
[Epoch 4/10] [Batch 424/469] D_loss: 0.2361 G_loss: 4.5502
Epoch 4 took 52.35 seconds
[Epoch 5/10] [Batch 55/469] D_loss: 0.5329 G_loss: 1.2562
[Epoch 5/10] [Batch 155/469] D_loss: 0.4543 G_loss: 3.1797
[Epoch 5/10] [Batch 255/469] D_loss: 0.3997 G_loss: 1.8319
[Epoch 5/10] [Batch 355/469] D_loss: 0.1676 G_loss: 3.1557
[Epoch 5/10] [Batch 455/469] D_loss: 0.2102 G_loss: 3.1613
Epoch 5 took 52.30 seconds
[Epoch 6/10] [Batch 86/469] D_loss: 0.4153 G_loss: 1.7600
[Epoch 6/10] [Batch 186/469] D_loss: 0.6820 G_loss: 2.2638
[Epoch 6/10] [Batch 286/469] D_loss: 0.5239 G_loss: 2.5355
```

Q&A

Q: 生成器和鉴别器的Loss是不是越低越好？为什么？

A: 不一定。若判别器Loss过低（比如说接近0），而生成器却还没充分收敛，就会因为判别器过于强大导致生成器无法学习，甚至陷入**模式崩溃**；反之亦然。理想的对抗训练过程应该两者势均力敌，**二者Loss存在动态平衡**，不是任一方Loss变低了就一定有利于训练，而是要两边的Loss在博弈中**共同震荡收敛**。

Q: 在本代码中，生成的数字是无法控制的（即采样一个噪声，把噪声输入到生成器中，得到的数字无法预测是什么），简单描述一下如何才能生成可控的数字（即想生成哪个数字就生成哪个数字）？

A: 可以采用**条件生成对抗网络（CGAN）**，令判别器对图片上的数字进行标签分类，把分类结果的反馈加入生成器损失函数。